

SYSTEM ANALYSIS AND DESIGN (SAD) COMPLETE NOTES

UNIT 1: SYSTEM DEVELOPMENT FUNDAMENTALS

1. System Definition

A system is an organized set of interrelated components that work together to achieve objectives by processing input into output.

Components:

- Inputs
- Processes
- Outputs
- Feedback
- Control
- Boundary
- Environment
- Interfaces

Characteristics:

- Organization
- Interaction
- Interdependence
- Integration
- Goal-oriented
- Adaptability
- Holism

Types: Open/Closed, Physical/Conceptual, Static/Dynamic.

CASE Tools:

CASE tools automate SDLC activities.

Types:

- Upper CASE: analysis/design
- Lower CASE: coding/testing
- Integrated CASE: full lifecycle

Modern SAD:

- Agile

- RAD

- OOAD

- Prototyping

- SOA

SDLC:

Planning → Analysis → Design → Implementation → Testing → Deployment → Maintenance

Waterfall Model:

Sequential and rigid model.

SOA:

Service-based architecture with reusable components.

System Acquisition:

Build, buy, outsource systems.

Reuse: using existing components.

Project Management:

Gantt Chart, PERT, WBS.

UNIT 2: PLANNING

Feasibility Analysis:

- Technical
- Economic
- Operational
- Schedule
- Legal/Ethical

Techniques:

NPV, ROI, Payback Period

$$NPV = \sum (CF / (1+r)^t) - \text{Initial Cost}$$

$$ROI = (\text{Profit/Cost}) * 100$$

PERT:

$$\text{Expected time} = (O + 4M + P) / 6$$

Planning:

Top-down vs Bottom-up

Baseline Project Plan:

Approved project document.

UNIT 3: ANALYSIS

Requirements Elicitation:

Interview, Observation, Questionnaire, Document analysis, JAD

DFD:

Context → Level 0 → Level 1

ER Model:

Entity, Attribute, Relationship

Business Rules:

Constraints for system behavior

Data Modeling:

Logical and conceptual design

Decision Table/Tree:

Business logic representation

UNIT 4: DESIGN**Normalization:**

1NF: atomic

2NF: no partial dependency

3NF: no transitive dependency

BCNF: every determinant is candidate key

ER to Relational Mapping:

Entity $\rightarrow$ Table

1:M $\rightarrow$ Foreign key

M:N $\rightarrow$ Junction table

Database Design:

Relational model uses tables.

UI Design:

Forms, menus, reports, interfaces

UNIT 5: IMPLEMENTATION & MAINTENANCE

Documentation:

System vs User docs

Maintenance:

Corrective, Adaptive, Perfective, Preventive

Testing:

Unit, Integration, System, Acceptance

Implementation:

Installation, training, support

Why Projects Fail:

Poor planning, unclear requirements, lack of user involvement, budget issues.